How to Prepare Input Data for MetaboReport?

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Introduction

MetaboReport provides step-by-step guidance on each page to help you perform a data analysis.

MetaboReport accepts a variety of outputs produced by different software tools, such as **XCMS**, **MSDIAL**, and **Compound Discoverer**. The output table can be submitted to MetaboReport with very little modification.

MetaboReport requires two types of information: **peak-area data** and **sample metadata**.

> `Peak area`: the integrated area under a chromatographic peak. It is commonly used as a relative measure of feature abundance. Depending on the data-processing software, peak intensity or peak height may be reported instead.
>
> `Sample metadata`: labels or group information that describe each sample.

There are **two ways** to prepare input data for MetaboReport:

1. Peak-area data and metadata are stored in the same table. Upload one table.

2. Peak-area data and metadata are stored in separate tables. Upload both tables.

This manual explains how to prepare both input formats.

Data Preparation

Input tables can be CSV (`.csv`) or Microsoft Excel (`.xls` or `.xlsx`) files.

Demo Data

The following example shows how to prepare an input table for MetaboReport.

Suppose you want to compare metabolic profiles between wild-type (**WT**) and mutant (**MU**) mice. Each biological sample is therefore labeled as either **WT** or **MU**.

Suppose sex is also relevant to the study, so a second metadata field records each biological sample as `Male` or `Female`.

Pooled QC (`QC`) samples are included to monitor analytical variation.

This study contains two metadata fields: Group 1 contains WT, MU, and QC; Group 2 contains Male, Female, and QC. Each sample therefore has two labels. For example, Sample_3 is WT and Male, whereas Sample_13 is QC in both fields (Table 1).

Table 1. Experiment design of a metabolomics study.

Sample	Sample Label 1	Sample Label 2
Sample_1	WT	Male
Sample_2	WT	Female
Sample_3	WT	Male
...	...	...
Sample_12	MU	Female
Sample_13	QC	QC
Sample_16	QC	QC

The study contains 16 samples (Table 1). After LC-MS analysis and data processing, you may obtain a table like Table 2. Each row represents a mass feature, and the columns contain feature descriptors or sample peak areas.

Table 2. An example of sample-feature table.

mz	Retention Time	Identification	PeakArea of Sample_1	PeakArea of Sample_2	....	PeakArea of Sample_16
133.014	3.23	Malic acid	123000	120000	...	801000
144.012	2.44	Unknown	678555	666000	...	1200
333.331	10.18	Unknown	3400	2000	...	560000
234.446	5.52	Unknown	45000	55000	...	46000
...	...	...	...	...	...	...
1034.552	25.33	Tomatine	4780	4300	...	12000

Table 2 must be modified to create an input table for MetaboReport.

As described above, there are two supported input formats.

Method 1: Peak-area data and metadata in one table

For Method 1, rename all peak-area columns shown in Table 2.

For example, rename `PeakArea of Sample_1` to `Sample_WT_Male_1`. See Table 3.

If the sample-feature table was generated by Compound Discoverer, `PeakArea of Sample_1` may instead be renamed `WT_Male_1`. The **Sample** prefix is optional for the Compound Discoverer format.

Explanation:

- Add the case-sensitive prefix `Sample_` to every peak-area column so MetaboReport can recognize it.
- Sample_1 has two metadata values (Table 1), so **WT** and **Male** follow `Sample_` and are separated by underscores. Metadata values are case-sensitive; for example, **WT** and **wt** are treated as different groups.
- Add a unique number after the metadata to distinguish sample replicates.

Table 3. Sample-Feature table for MetaboReport prepared by method 1.

mz	Sample_WT_Male_1	Sample_WT_Female_2	Sample_QC_QC_16
133.014	123000	120000	801000
144.012	678555	666000	1200

Note: You may apply this naming convention to raw-data filenames before data processing so that the resulting sample-feature table can be used directly by MetaboReport.

Method 2: Peak-area data and metadata in separate tables

Method 2 requires a separate metadata table.

For **peak area table**:

- Name every peak-area column using the case-sensitive format `Sample_<number>` so MetaboReport can recognize it (for example, `Sample_1`, `Sample_2`).
- Other columns are optional and may remain in the table.
- The peak-area table may also use the full naming convention described in Method 1.

See Table 4 for details.

Table 4. Peak area table for MetaboReport prepared by method 2.

mz	Retention Time	Identification	Sample_1	Sample_2	....	Sample_16
133.014	3.23	Malic acid	123000	120000	...	801000
144.012	2.44	Unknown	678555	666000	...	1200
333.331	10.18	Unknown	3400	2000	...	560000
234.446	5.52	Unknown	45000	55000	...	46000
...	...	...	...	...	...	...
1034.552	25.33	Tomatine	4780	4300	...	12000

For **Metadata table**:

Prepare a separate table that describes the metadata for each sample (Table 5). Column headers may use any names, but the first column must contain sample identifiers. Metadata values are case-sensitive, so use them consistently.

Table 5. Metadata table for MetaboReport prepared by method 2.

Sample	Sample Label 1	Sample Label 2
Sample_1	WT	Male
Sample_2	WT	Female
Sample_3	WT	Male
...	...	...
Sample_12	MU	Female
Sample_13	QC	QC
Sample_16	QC	QC

Important: MetaboReport currently assigns metadata by row order. Use the same `Sample_<number>` identifiers in Tables 4 and 5, and list them in exactly the same order. Otherwise, metadata will be assigned to the wrong samples.

Demo Data

Two demo datasets, prepared using Methods 1 and 2, are available from the **Home** page of the MetaboReport app.

Use these demo datasets as references when preparing your own data.

Common Mistakes

1. My data table is not accepted or not recognized by MetaboReport.

Most import errors result from an unsupported input format. Check the following points when preparing a data table:

- Metadata values are case-sensitive. For example, **WT**, **Wt**, and **wt** are treated as different groups, so use labels consistently.
- `Sample_` is the reserved, case-sensitive prefix used by MetaboReport to recognize **peak-area** columns. For Method 2, use the exact format `Sample_<number>`.
- **QC** is a reserved keyword used only for QC samples.
- Avoid unintended **white space**. `WT` and WT are treated as different metadata values.
- For tables generated by **Compound Discoverer**, export all `Area` columns and select `Compound Discoverer (CD)` on the **Upload Data** page. For tables produced by other software, select `Other format`.

2. It takes too long to generate the report.

- Additional calculations are performed during report generation, so the process may take a few minutes.
- If report generation fails, restart the app session and rerun the analysis.